

R V College of Engineering

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

IV Semester B.E Regular/Supplementary Examinations - June/July-2025.

Electronics and Communication Engineering

Course : Transmission lines and electromagnetic fields-EC345AT

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

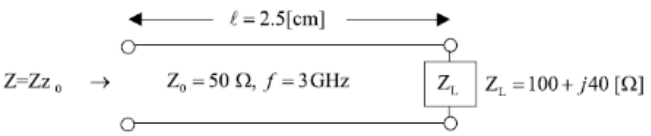
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

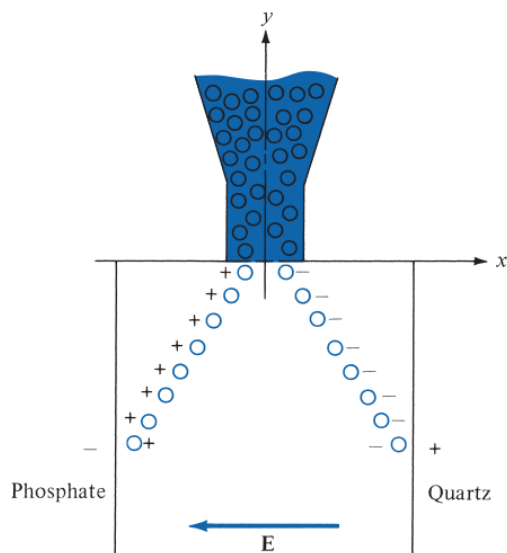
Part A

Question No	Question	M	CO	BT
1.1	A $120\ \Omega$ lossless transmission line is terminated at a load impedance $Z_L = 200 - j240\ \Omega$. Find the reflection coefficient Γ .	01	1	2
1.2	A distortion less line has $R_1 = 6\ \Omega\ km^{-1}$, $G = 3.2\ \Omega^{-1}\ km^{-1}$ and $L = 8.847\ \mu H\ km^{-1}$. Find attenuation constant & line capacitance of the line	02	1	3
1.3	What is the distance between two consecutive voltage (or current) maxima or minima on a lossless transmission line?	01	1	2
1.4	Design a single-section quarter-wave matching transformer to match a $120\ \Omega$ load to a $50\ \Omega$ transmission line at $f_0 = 3\ GHz$. Find the required characteristic impedance of the matching section.	01	2	2
1.5	Given: $\vec{D} = 8\rho \sin \phi \hat{\rho} + 4\rho \cos \phi \hat{\phi}\ \mu C/m^2$. Find the volume charge density ρ_v at point $P(5, 30^\circ, -4)$	02	2	3
1.6	Plane $z = 10\ m$ carries charge $20\ nC/m^2$. The electric field intensity at the origin is _____	01	1	2
1.7	Two point Charges $-4\ \mu C$ & $5\ \mu C$ are located at $(2, -1, 3)$ and $(0, 4, -2)$ respectively. Find the potential at $(1, 0, 1)$ assuming zero potential at infinity.	02	1	3
1.8	An electric potential field is produced by point charges $1\ \mu C$ and $4\ \mu C$ are located at $(-2, 1, 5)$ and $(1, 3, -1)$, respectively. Find the energy stored in the field.	02	2	3
1.9	Current sheets of $20\hat{a}_x\ A/m$ and $-20\hat{a}_x\ A/m$ are located at $y = 1$ and $y = -1$ respectively. Find $\vec{H}$ in region $-1 < y < 1$.	02	3	2
1.10	Compute the value of k , for the vector field $4x\hat{a}_x + 2ky\hat{a}_y + 8\hat{a}_z$ to represent a magnetic field.	02	3	3
1.11	An airplane with a metallic wing of span $36\ m$ flies at $410\ m/s$ in a region where the vertical component of the earth's magnetic field is $0.4\ mWb/m^2$. Find the emf induced on the airplane wing.	01	4	2
1.12	In a certain medium, $\vec{E} = 10 \cos(10^8 t - 3y)\hat{a}_x\ V/m$. What type of medium is it?	01	4	4
1.13	The electric field intensity of a plane wave traveling in free space is given by the following expression $\vec{E}(x, t) = 24\pi \cos(\omega t - k_0 x)\hat{a}_y\ V/m$. In this field,	02	4	3

consider a square area $10 \text{ cm} \times 10 \text{ cm}$ on a plane $x + y = 1$. Compute the time-averaged power passing through the area.

Part B

Question No	Question	M	CO	BT
2a	A telephone line has $R = 30 \Omega/\text{km}$, $L = 100 \text{ mH}/\text{km}$, $G = 0$, and $C = 20 \mu\text{F}/\text{km}$. At $f = 1 \text{ kHz}$, obtain: (i) The characteristic impedance of the line. (ii) The propagation constant. (iii) The phase velocity.	08	2	3
2b	A distortionless transmission line satisfies the condition $RC = LG$. Assume the line operates at 80 MHz. Given the line parameters: - Resistance $R = 10 \text{ m}\Omega/\text{m}$ - Capacitance $C = 82 \text{ pF}/\text{m}$ - Inductance $L = 0.6 \mu\text{H}/\text{m}$ Calculate: (i) The characteristic impedance Z_0 (ii) The propagation constant γ	08	2	3
3a	A cube is defined by the region $1 < x, y, z < 1.2$. The electric flux density is given by $\vec{D} = 2x^2y\hat{a}_x + 3x^2y^2\hat{a}_y \quad \text{mC}/\text{m}^2.$ Find the total electric flux leaving the closed surface of the cube.	06	2	3
3b	A charge distribution in free space has $\rho_v = 2r \text{ nC}/\text{m}^3$ for $0 \leq r \leq 10 \text{ m}$ and zero otherwise. Determine E at $r = 2 \text{ m}$ and $r = 12 \text{ m}$.	06	3	3
3c	Explain briefly about Single-Stub Tuner (Matching) with necessary diagrams.	04	3	1
OR				
4a	The air-filled two-wire line has a characteristic impedance of 50Ω and is operated at $f = 3 \text{ GHz}$. The load is $Z_L = 100 + j40 \Omega$.  (i) For the line above, find Z_L on the chart. (ii) What is the line impedance 2.5 cm from the load? (iii) What is the VSWR on the line?	06	2	3
4b	A practical application of electrostatics is in electrostatic separation of solids. For example, Florida phosphate ore, consisting of small particles of quartz and phosphate rock, can be separated into its components by applying a uniform electric field as in Figure . Assuming zero initial velocity and displacement, determine the separation between the particles after falling 80 cm. Take $E = 500 \text{ kV}/\text{m}$ and $Q/m = 9 \mu\text{C}/\text{kg}$ for both positively and negatively charged particles.	10	3	3



5a

Given a surface charge density of 8 nC/m^2 on the plane $x = 2$, a line charge density of 30 nC/m is positioned at $x = 1, y = 2$, and a $1 \text{ } \mu\text{C}$ point charge at $P(-1, -1, 2)$. Find V_{AB} for points $A(3, 4, 0)$ and $B(4, 0, 1)$.

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5b

The surface $x = 0$ separates two perfect dielectrics. For $x > 0$, let $\epsilon_1 = 3$, while $\epsilon_2 = 5$ where $x < 0$. If $\vec{E}_1 = 80\hat{a}_x - 60\hat{a}_y - 30\hat{a}_z \text{ V/m}$, find:

(i) $\vec{D}_2$ and $\vec{E}_2$.

(ii) The angles $\vec{E}_1$ and $\vec{E}_2$ make with the interface.

(iii) Energy densities in each medium.

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OR

6a

Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z \geq 0$, $\epsilon_{r1} = 4$ and for $z < 0$, $\epsilon_{r2} = 3$. A uniform electric field $\vec{E}_1 = 5\hat{a}_x - 2\hat{a}_y + 3\hat{a}_z \text{ KV/m}$ exists for $z \geq 0$. Find

(i) $\vec{E}_2$ for $z \leq 0$

(ii) The angles $\vec{E}_1$ and $\vec{E}_2$ make with the interface

(iii) The energy densities in J/m^3 in both dielectrics.

(iv) The energy within a cube of side 2 m centered at $(3, 4, -5)$.

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6b

Conducting cylinders at $\rho = 2 \text{ cm}$ and $\rho = 8 \text{ cm}$ in free space are held at potentials of 60 mV and -30 mV , respectively. Using Laplace equation,

(i) Derive and determine V and $\vec{E}$ in the region between the shells.

(ii) Determine the maximum $\vec{E}$ if the region has $\epsilon_r = 2.5$.

(iii) Determine the surface where $V = 30 \text{ mV}$.

(iv) The capacitance of the capacitor.

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7a

Two current elements $I_1 dl_1 = 4 \times 10^{-5} \hat{a}_x \text{ A.m}$ at $(0, 0, 0)$ and $I_2 dl_2 = 6 \times 10^{-5} \hat{a}_y \text{ A.m}$ at $(0, 0, 1)$ are in free space. Find $\vec{H}$ at $(3, 1, -2)$.

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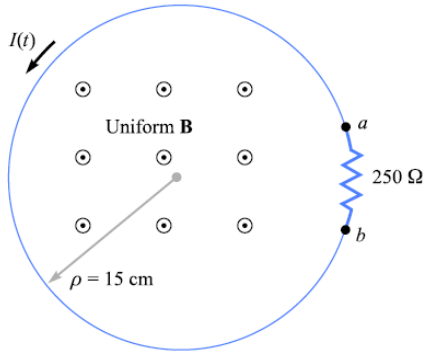
7b

The magnetic vector potential of a current distribution in free space is given by $\vec{A} = 15e^{-\rho} \sin \phi \hat{a}_z \text{ Wb/m}$. Find $\vec{H}$ at $(3, \pi/4, -10)$. Calculate the

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	flux through $\rho = 5, 0 \leq \phi \leq \frac{\pi}{2}, 0 \leq z \leq 10$.			
OR				
8a	Line $x = 0, y = 0, 0 \leq z \leq 10$ m carries current 2 A along $\hat{a}_z$. Calculate $\vec{H}$ at points (i) (5, 0, 0) (ii) (5, 15, 0) (iii) (5, 5, 0) (iv) (5, -15, 0)	08	2	2
8b	Derive the boundary conditions for magnetic media and explain with suitable diagram.	08	2	2
9a	The electric field intensity in the region $0 < x < 5$ m, $0 < y < \pi/12$ m, $0 < z < 6$ m in free space is given by $\vec{E} = C \sin(12y) \sin(az) \cos 2 \times 10^{10}t \hat{a}_x$ V/m. Beginning with the $\nabla \times \vec{E}$ relationship, use Maxwell's equations to find a numerical value for a , if it is known that a is greater than zero.	08	4	4
9b	Region 1 is a lossless medium for which $y \geq 0, \mu = \mu_0, \varepsilon = 4\varepsilon_0$, whereas region 2 is free space, $y \leq 0$. If a plane wave $\vec{E}_i = 5 \cos(10^8t + \beta y) \hat{a}_z$ V/m exists in region 1, find (i) the total electric field component of the wave in region 1, (ii) the time-average Poynting vector in region 1, (iii) the time-average Poynting vector in region 2.	08	4	3
OR				
10a	In a lossless medium ($\varepsilon_r = 4.5, \mu_r = 1$), a uniform plane wave $\vec{E} = 8 \cos(\omega t - \beta z) \hat{a}_x - 6 \sin(\omega t - \beta z) \hat{a}_y$ V/m propagates at 40 MHz. (i) Find $\vec{H}$. (b) Determine β, λ, η and velocity of the wave.	10	4	3
10b	In the Figure below, let $\vec{B} = 0.2 \cos 120\pi t$ T, and assume that the conductor joining the two ends of the resistor is perfect. It may be assumed that the magnetic field produced by $I(t)$ is negligible. Find (i) $V_{ab}(t)$ (ii) $I(t)$. 	06	4	3

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IV Semester BE Regular/Supplementary Examinations June/July - 2025.

Course : Maths for communication-EC241AT

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	A random process has a sample function of the form $X(t) = \begin{cases} A, & 0 \leq t \leq 1 \\ 0, & \text{Elsewhere} \end{cases}$ Where A is a random variable that is uniformly distributed from 0 to 10. The autocorrelation of the process is _____.	02	1	2
1.2	For two zero-mean discrete WSS processes $X[n]$, $Y[n]$, the cross-correlation is: $R_{XY}[k] = 0.5 * (0.9)^{ k }$. Find $\text{Cov}[X[n], Y[n+3]]$.	02	4	2
1.3	A signal $x_1(t)$ is band-limited to 5 kHz. What is the Nyquist sampling rate for $x_1(t) * x_1(t)$?	01	1	2
1.4	A signal having bandwidth 4MHz is transmitted by binary PCM, using 8bits per sample and a sampling rate of 10MHz. The number of quantization levels and the transmission bit rate are ____ and ____ respectively.	02	3	3
1.5	Given the signal $m(t) = 20(\cos 2000\pi t)(\cos 8000\pi t)$, what is the minimum sampling rate based on the band pass sampling theorem?	01	1	1
1.6	Two devices that exchange data between them have settable clock rates. Case 1 – widely differing rates and Case 2 – Nominally same (but not identical). Multiplexing scheme employed for Case1=_____and for Case2=_____.	02	1	1
1.7	Number of basis functions required to represent Orthogonal	02	1	2

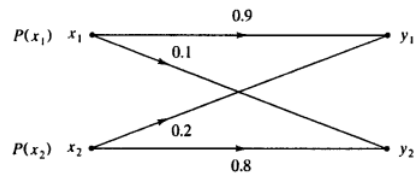
	signals=_____and Antipodal Signals=_____.			
1.8	A speech signal is sampled at a rate of 8 KHz and then encoded using 8bits/sample. The resulting binary data is then transmitted through an AWGN channel via 4-FSK. The bandwidth required for transmission=_____KHz.	02	1	2
1.9	How to achieve same error performance of binary DPSK in non-coherent binary FSK? Mention the applications of DPSK and non-coherent FSK.	02	2	2
1.10	A BPSK signal is used for transmitting information over an AWGN with a Power Spectral Density (PSD) of $N_0/2 = 10^{-10} \text{ W/Hz.}$ The transmitted signal energy is given by $E_b = (A^2 * T) / 2, \text{ where 'T' is the bit interval and 'A' is the signal amplitude.}$ Determine the signal amplitude (in mV) required to achieve an error probability of 10^{-6}, if the data rate is 10 kbps.	02	1	2
1.11	In center point sampling detection two signals with same probability of occurrence are defined as $S_1(t) = A g(t)$ $S_2(t) = -\frac{A}{2} g(t)$ Here $g(t)$ is a rectangular pulse with duration T. Find Probability of error if $A = 4V$ and $\frac{N_0}{2} = 600 \times 10^{-3} \text{ W/Hz.}$	02	1	2

Part B

Question No	Question	M	CO	BT
2a	A modulated random signal $Y(t) = A X(t) \cos(\omega_c t + \theta)$, where $X(t)$ is the random message signal and is a zero-mean stationary random process. The carrier amplitude A and the frequency ω_c are constants and phase θ is a random variable uniformly distributed over $[-\pi, \pi]$. Assuming that $X(t)$ and θ are independent, check if $Y(t)$ is an ergodic random process and WSS.	08	1	3
2b	Consider a binary channel shown in Fig. (a) Find the channel matrix of the channel. (b) Find $P(y_1)$ and $P(y_2)$ when $P(x_1) = P(x_2) = 0.5$. (c) Find the joint probabilities $P(x_1, y_2)$ and $P(x_2, y_1)$ when $P(x_1) = P(x_2) =$	08	1	3

0.5.

(d) Find the probabilities $P(x_1/y_2)$ and $P(x_2/y_1)$



In a single-integration Delta Modulation System, the voice signal is sampled at a rate of 64 kHz, similar to PCM system. The maximum signal amplitude is normalized as $A_{max} = 1$.

i) Determine the minimum value of step size to avoid overload.

ii) Determine the granular noise power if the voice signal bandwidth is 3.4 kHz.

iii) Assuming that the voice signal is sinusoidal, determine the signal power and SNR.

iv) Assuming that the voice signal amplitude is uniformly distributed in the range $(-1,1)$, compute the signal power and SNR.

v) Calculate the minimum transmission bandwidth.

A waveform, $x(t)=10\cos(1000t+\pi/3)+20\cos(2000t+\pi/6)$ is to be uniformly sampled for digital transmission.

(i) What is the maximum allowable time interval between sample values that will ensure perfect signal reproduction?

(ii) If we want to reproduce 1 hour of this waveform, how many sample values need to be stored?

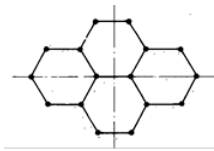
OR

In a binary PCM system, the output signal-to-quantizing-noise ratio is to be held to a minimum of 40 dB. Determine the number of required levels, and find the corresponding output signal-to-quantizing-noise ratio.

With a block diagram, discuss the working of the following in detail: Differential Pulse Code Modulation and Demodulation & Delta Modulation and Demodulation.

Four signals m_1 to m_4 comprising m_1 of BW 3.6KHz and 10 bits/sample, and m_2, m_3, m_4 Each BW= 1.2KHz and 12 bits /sample are to be multiplexed. Devise a multiplexing scheme and explain how to perform frame recovery at the demultiplexer. Assuming header of size 8 bits.

5b	The set of 4 signals are described as follows: $s_1(t) = \{ A, 0 \leq t$ $s_2(t) = \{ A, 0 \leq t$ $s_3(t) = -s_1(t)$, $s_4(t) = -s_2(t)$. Draw the constellation diagram and express the signals in terms of basis functions.	08	3	4
OR				
6a	With the help of Finite State Machine, describe the principle of operation frame identification of T1 frame implemented at receivers for Quasi-synchronous digital de-multiplexing scheme	08	3	2
6b	Consider the signals $s_1(t)$, $s_2(t)$, $s_3(t)$ and $s_4(t)$ as shown in figure. Use Gram Schmidt orthogonalization procedure to find an orthonormal basis functions for the set of signals. Express these signals in terms of $s_1(t)$ $s_2(t)$ $s_3(t)$ $s_4(t)$ basis functions.	08	3	4
7a	With relevant mathematical equations, discuss the transmitter and receiver architecture of 8-PSK. Deduce the expression for PSD.	08	2	3
7b	Obtain the PSD of a unipolar signaling scheme. What are the parameters obtained from PSD?	08	2	3
OR				
8a	With relevant mathematical equations, discuss the transmitter and receiver architecture of BPSK. Deduce the expression for PSD.	08	2	3
8b	With relevant mathematical equations, discuss the transmitter and receiver architecture of BFSK (Coherent and non-coherent receiver modes)	08	2	3
9a	How does maximizing Aposteriori or Likelihood reduces the probability of error. Justify the answer.	08	3	3
9b	For the 16-QAM constellation of Fig, the distance between any two adjacent points is 'A'. Obtain expression for a union bounded optimistic and pessimistic probability of error.	08	3	3



OR

10a

Obtain the probability of error expression in terms of d_{min} . From the obtained expression find the probability of error for Bipolar and Unipolar line codes

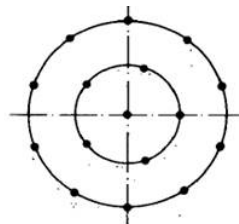
08

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10b

Compute the average energy for the given constellation as shown in fig, assuming distance between adjacent symbols is 'A'. Also find the probability of error for optimistic and pessimistic case.



08

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IV Semester B.E Regular/Supplementary Examinations June/July-2025.

Common to EC/EI

Course : Signals and Systems-EC244AI

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

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Part A

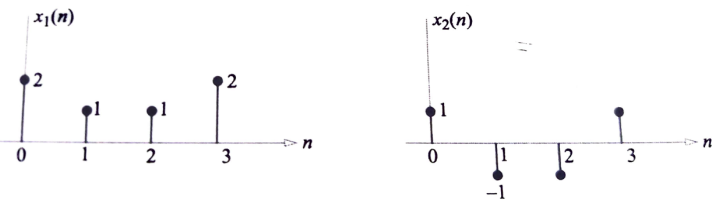
Question No	Question	M	CO	BT
1.1	Consider a discrete-time signal $x[n] = \sin(\pi^2 n)$, n being integer. Find whether the signal is periodic or not. If periodic, find its period.	02	2	4
1.2	A continuous-time signal $x(t) = 4 \cos(200\pi t) + 8 \cos(400\pi t)$, where t is in seconds, is input to a linear time-invariant (LTI) filter with the impulse response $h(t) = \begin{cases} \frac{2 \sin(300\pi t)}{\pi t}, & t \neq 0 \\ 600, & t = 0 \end{cases}$ Let $y(t)$ be the output of this filter. Compute the maximum value of $y(t)$.	02	1	3
1.3	Determine the step response of the system if the overall impulse response $h[n]$ is the combination of $h_1[n]$ and $h_2[n]$ placed in cascade connection, where $h_1[n] = 2\delta[n] + 4\delta[n - 2]$ and $h_2[n] = 3\delta[n + 1] + 5\delta[n - 1]$.	02	2	2
1.4	The two responses of an LTI system are given as $h_1(t) = e^{-2t}u(t)$ and $h_2(t) = e^{-t}u(t)$. Find the overall response $h(t)$ if the two systems are placed in cascade connection.	02	2	2

1.5	Evaluate the DTFS representation of the signal: $x(n) = \cos\left(\frac{6\pi n}{17} + \frac{\pi}{3}\right)$.	02	3	2
1.6	Let $x(n) = (1/2)^n u(n)$, $y(n) = x^2(n)$ and $Y(e^{j\Omega})$ to be the Fourier transform of $y(n)$ then find $Y(e^{j0})$.	02	3	2
1.7	DFT of a signals is $\cos^2 \Omega + \sin^2 2\Omega$. Determine the signal corresponding to the DFT.	02	3	2
1.8	In radix-2 FFT, if the number of points $N = 16$, then the number of complex additions and number of complex multiplications are respectively _____ and _____.	02	3	1
1.9	Discuss the impact of Zero Crossing Rate (ZCR) in signal processing techniques.	02	3	2
1.10	What is spectral leakage and how does it affect feature extraction?	02	4	1

Part B

Question No	Question	M	CO	BT
2a	Determine whether the following signals are energy signals, power signals, or neither. (i) $x(t) = e^{-at}u(t)$, $a > 0$ (ii) $x(t) = tu(t)$ (iii) $x[n] = (-0.5)^n u[n]$ (iv) $x[n] = 2e^{j3n}$	08	1	2
2b	A discrete-time system is both linear and time-invariant. Suppose the output due to an input $x[n] = \delta[n]$ is given in the Figure (a) below. (i) Find the output due to an input $x[n] = \delta[n - 1]$. (ii) Find the output due to an input $x[n] = 2\delta[n] - \delta[n - 2]$. (iii) Find the output due the input depicted in the Figure (b).	08	1	3

	(a) (b)			
3a	For the given impulse response $h[n] = u[n+1] - u[n-4]$, the LTI system is excited by the input signal $x[n] = u[n] - 2u[n-2] + u[n-4]$. Obtain the output response $y[n]$ and plot the same. Use graphical method to solve the problem.	10	2	4
3b	Find the step response of the system where the impulse response is given by $h(t) = tu(t)$. Plot the step response.	06	2	3
OR				
4a	Compute the convolution integral: $u(t+1) * u(t-2)$. Plot the output response.	06	2	3
4b	Evaluate the following discrete time convolution sum analytically and plot $y[n]$, where, $y[n] = \beta^n u[n] * u[n-3]$, $ \beta < 1$.	10	2	2
5a	Find the DTFS and DTFT representations of the following periodic signal. Sketch the magnitude and phase spectrum. $x[n] = 1 + \sum_{m=-\infty}^{\infty} \cos\left(\frac{\pi m}{2}\right) \delta[n-m]$	08	3	2
5b	The input to a discrete time LTI system is $x(n) = \cos\left(\frac{\pi n}{8}\right) + \sin\left(\frac{3\pi n}{4}\right)$. Use the DTFT to find the output of the system, $y(n)$, if the impulse response is $h(n) = \frac{\sin(\frac{\pi n}{4})}{\pi n}$. Also sketch $X(e^{j\Omega})$, $H(e^{j\Omega})$ and $Y(e^{j\Omega})$.	08	3	2
OR				
6a	Consider the system depicted in Figure. The input to a discrete time system is $x(n) = \delta(n) - \frac{\sin\left(\frac{\pi}{4}n\right)}{\pi n}$, $w(n) = (-1)^n$ and the impulse response $h(n) = \frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}$.	10	4	4

	Use the DTFT to determine the output $y(n)$. Also sketch $G(e^{j\Omega})$, $X(e^{j\Omega})$, $H(e^{j\Omega})$ and $Y(e^{j\Omega})$.			
6b	Find the DTFS and DTFT representation for the following periodic signals, $x(n) = \cos \left[\frac{\pi}{8} n \right] + \sin \left[\frac{\pi}{5} n \right]$	06	3	2
7a	For the sequences, $x_1(n)$ and $x_2(n)$ shown below:  (i) Compute the circular convolution of these sequences for $N = 4$ and $N = 8$. (ii) Compute the linear convolution of these sequences. (iii) What value of N is necessary so that linear and circular convolution are the same on the N-point interval? (iv) Determine the non-zero lengths of $x_1(n)$ and $x_2(n)$ and justify how could have found the results of part (iii) without performing the calculations.	08	3	4
7b	Assume that a complex multiplication takes $1 \mu s$ and that the amount of time take to compute N-point DFT is determined by the amount of time it takes to perform all of the multiplications. (i) How much time it takes to compute a 64-point DFT directly? (ii) How much time is required if an FFT is used? (iii) Repeat parts (i) and (ii) for a 256-point DFT.	08	3	1
OR				
8a	Find the 4-point circular convolution of $x(n)$ and $h(n)$ as given below: $x(n) = \{1, 1, 1, 1\}$ and $h(n) = \{1, 0, 1, 0\}$ using radix-2 DIF-FFT algorithm.	10	3	3
8b	Two sequences $[a, b, c]$ and $[A, B, C]$ are related as,	06	4	4

	$\begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^{-1} & W_3^{-2} \\ 1 & W_3^{-2} & W_3^{-4} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \text{ where, } W_3 = e^{j\frac{2\pi}{3}}.$ If another sequence $[p, q, r]$ is derived as, $\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^1 & W_3^2 \\ 1 & W_3^2 & W_3^4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & W_3^2 & 0 \\ 0 & 0 & W_3^4 \end{bmatrix} \begin{bmatrix} A/3 \\ B/3 \\ C/3 \end{bmatrix},$ then establish a relationship between the sequences $[p, q, r]$ and $[a, b, c]$.			
9a	A biomedical research team is developing an intelligent system to assist neurologists in diagnosing epilepsy. The system uses EEG (electroencephalogram) signals, which record brain activity through electrodes placed on the scalp. Since seizures are often reflected as sudden changes in signal characteristics, the team is working on extracting meaningful features from both time and frequency domains to train a classifier that can distinguish between normal and seizure episodes. (i) Explain how energy and skewness in the time domain could indicate seizure activity. (ii) Why might frequency domain features be better suited for detecting certain patterns in EEG data? (iii) How can combining both types of features improve the accuracy of seizure classification?	10	4	4
9b	Discuss how a spectrogram is derived using the Short-time Fourier transform.	06	2	3
OR				
10a	A tech startup is working on enhancing user experience in smart virtual assistants by recognizing the speaker's emotional state. They collect audio recordings of different users speaking under various emotional conditions such as happiness, sadness, anger, and surprise. By analyzing speech signals for patterns in both time and frequency, the team aims to extract features that reflect pitch, energy, and voice modulation, which can then be used to classify emotions in real time. (i) What role does the envelope of the signal play in identifying emotional states? (ii) Discuss how dominant frequency and peak values can be indicators of vocal pitch and intensity for emotion detection. (iii) Propose a basic pipeline involving feature extraction and classification for speech emotion recognition.	10	4	4

10b	Discuss the importance of mean, variance, and standard deviation in time-domain feature extraction. How do these features help in distinguishing between different types of signals? Provide an example.	06	2	3
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